

Commitment, Flexibility, and Representation

The Institutional Politics of Climate Compensation

Comparative Political Economy Seminar

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Outline

Motivation

Theoretical framework

Empirical design

Conclusion and outlook

Motivation

Two “stylised facts” about climate public opinion:

- Public support for ambitious climate policy is **broad, but shallow** – high generic support (Andre et al., 2024; Gazmararian, Mildemberger, and Tingley, 2025), but limited willingness to pay (Dechezleprêtre et al., 2025) – and **deservingness** (“fairness”) matters (Sommer, Mattauch, and Pahle, 2022).
- Risk of climate backlash as a result of heightened **unemployment risk** in carbon-intensive sectors (Heddesheimer, Hilbig, and Voeten, 2025), **rising energy prices** (Konc et al., 2024; Voeten, 2025), and **broadier distributional effects** of climate policy instruments like bans (Colantone et al., 2023), standards, and carbon taxes/prices (Furceri, Ganslmeier, and Ostry, 2023; Konradt and Mangiante, 2025).

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Climate compensation has gained popularity as a tool for stabilising and, potentially, even broadening the “climate coalition” (Edenhofer and Genovese, 2024).

- Buy the time necessary for long-term measures to yield tangible benefits → deal with “tragedy of the horizons” (Carney, 2015)

Situating our argument in the general climate literature

Evidence on political efficacy is **decidedly mixed** – both observationally and experimentally.

- Some evidence that compensation can “work” politically (Bolet, Green, and González-Eguino, 2024; Colantone et al., 2023; Jagers, Martinsson, and Matti, 2019; Woerner et al., 2024).
- But: it can also fail (Douenne and Fabre, 2022; Mildemberger et al., 2022; Stutzmann, 2025).
- Echoes work in other policy domains, like **trade** (M. Kim and Gulotty, 2024; S. E. Kim and Pelc, 2021) and **automation** (Falck et al., 2024; Gingrich, 2019).

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Most research examines **policy design**, e.g. the **generosity** (Schaffer, 2023), **scope** (Gaikwad, Genovese, and Tingley, 2022) or the other policies with which climate compensation is **bundled** (Baute, 2025; Bergquist, Mildemberger, and Stokes, 2020; Gazmararian, 2024; Wicki, Huber, and Bernauer, 2020).

Less attention has been paid to the **institutional context** – the *when/how/by whom* – in which compensation decisions are made, though notable exceptions exist (Gazmararian and Tingley, 2023, 2026).

Situating our argument in the (climate) institutional literature

Following seminal work in the political economy literature (Dixit, [1996](#); Kydland and Prescott, [1977](#); Rodrik, [1989](#)), Gazmararian and Tingley, [2023](#) highlight the need for **institutions** to address the **commitment problem** inherent in climate compensation.

A separate strand of the literature stresses the importance of “**stakeholder engagement**” (Im, [2024](#); Im et al., [2025](#); Nacke et al., [2024](#); Siderius, [2025](#)).

- Normative justifications aside, the political rationale for **representation** remains unclear. Recent work has started to examine this more systematically, but with different research questions in mind (Bayer and Genovese, [2025](#); Mandelli, [2025](#)).

A nascent, third body of work focuses on the role of **(local) state capacity** in impeding or enabling more stringent climate policy (Davidovic, [2025](#); Feld and Fetzer, [2024](#)).

- Important in light of EU Just Transition Plans

Gaps in the literature

Despite extensive work on climate policy support, two substantive **gaps** remain:

1. **Fragmented or “bundled” treatment of institutional features:** Existing research either examines these institutional features in isolation—focusing on the *credibility of promises* or on *stakeholder inclusion*—or it *bundles* them together.
2. **Unexplored individual-level trade-offs:** We know little about whether/how heterogeneous exposure to the net zero transition shapes variation in institutional preferences

Our contribution: We theorise and test how these two dimensions jointly shape citizens’ institutional preferences and their willingness to support ambitious climate policy.

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Building blocks

- Citizens differ not only in *how much* compensation they want, but in *how* compensation is governed and by *whom*.
- These preferences reflect beliefs about **credibility**, **efficiency**, and **deservingness considerations** in delivering compensation.

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- Citizens differ not only in *how much* compensation they want, but in *how* compensation is governed and by *whom*.
- These preferences reflect beliefs about **credibility**, **efficiency**, and **deservingness considerations** in delivering compensation.
- We distinguish three broad **exposure types**, which differ in the relative weights they attach to these considerations:
 1. **Present losers**: directly and currently affected by decarbonisation (e.g. workers or regions in carbon-intensive sectors).
 2. **Future losers**: anticipate exposure but with uncertainty about timing, magnitude, or sectoral scope.
 3. **Unaffected**: expect no direct material losses from the transition.
- **Caveat**: We abstract from **electoral accountability** (Martin, 2023) and **decentralisation** (Beramendi, 2012) → theoretical parsimony + similarity to real-world institutions

Two orthogonal institutional dimensions

We theorise two analytically distinct axes of institutional design for climate compensation:

1. **Temporal dimension:** *commitment* vs. *flexibility*

- Captures how stable compensation rules remain over time.
- High **commitment**: rules and allocations fixed in advance → enhances **credibility** but limits adaptability. High **flexibility** implies the reverse.
- Classic trade-off between **credibility** and **flexibility** (Amador, Werning, and Angeletos, 2006; Dixit, 1996).

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2. **Compositional dimension:** *technocratic* vs. *descriptively representative*

- Concerns **who** designs and implements compensation.
- **Technocratic** designs → insulated, expert decision-making; potentially enhances impartiality and efficiency, but risks **incongruence** with affected communities.
- **Descriptive representation** → includes representatives of affected constituencies; but risks **partiality**, favouritism, or capture.
- Citizens evaluate this trade-off between **impartiality** and **incongruence**.

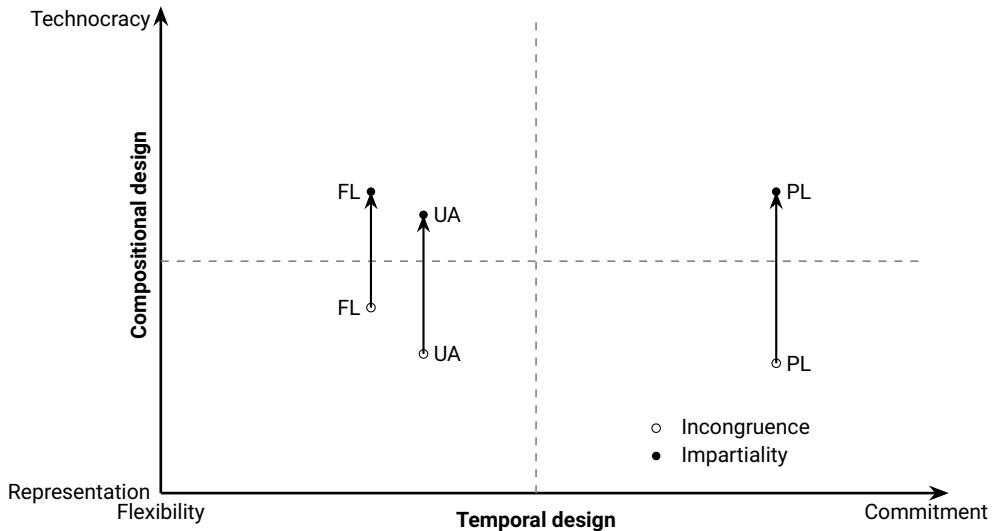
Institutional trade-offs and citizen preferences

- Along the **temporal dimension**:
 - Present losers → prefer **commitment** (credible protection now).
 - Future losers and the unaffected → prefer **flexibility** (adaptive response later).
- Along the **compositional dimension**:
 - Affected groups → prefer **representation**, insofar as this reduces **incongruence** between decision-makers and those affected.
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 - Concern about **partiality**, favouritism, or inefficiency → shifts preference toward **technocracy**.
 - Among the **unaffected**, the **efficiency–deservingness** calculus provides the evaluative lens through which they interpret the **impartiality–incongruence** trade-off:
 - Efficiency concerns ⇒ preference for **technocratic impartiality** (expert, rule-based, low risk of waste).
 - Deservingness concerns ⇒ preference for **representative congruence** (visible inclusion of those “who merit” compensation).
- Institutional designs that align with these group-specific preferences should increase **overall support for climate policy**.

Visualising the argument



Theoretical hypotheses

- H1 Exposure timing → Temporal design:** The more **present-skewed** an individual's climate-related losses, the stronger their preference for **commitment** over **flexibility**.
- H2 Exposure → Compositional design:** Those who perceive higher incongruence with decision-makers (often affected groups) prefer **representation** over **technocracy**
- H3 Impartiality concern:** Fear of partiality, favouritism, or inefficiency increases support for **technocratic** decision-making.
- H4 Evaluative logic among the unaffected:** Among those not directly exposed to losses, **efficiency** concerns lead to a preference for **technocracy**, while **deservingness** concerns favour **representation**; temporal preferences remain oriented toward **flexibility**.
- H5 Policy support:** Institutional designs aligning with citizens' preferences across both dimensions yield higher **support for climate policy**.

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Case selection

- The conjoint experiment will be conducted in **Germany, Spain, Czechia, and Poland**—four EU member states that maximise variation on:
 - **Exposure** to decarbonisation pressures and material transition risks;
 - **Experience** with compensation and stakeholder involvement;
 - **Administrative traditions** and perceptions of bureaucratic impartiality.
- Variation *across* and *within* countries enables tests of whether the theorised mechanisms generalise across contexts:
 - **Spain** and **Germany** – contrasting cases: Spain’s decentralised “Just Transition Agreements” boosted electoral support (Bolet, Green, and González-Eguino, [2024](#)), while Germany’s corporatist “Coal Commission” built elite consensus but proved less effective in preventing electoral backlash (Mijin Cha, [2024](#); Stutzmann, [2025](#));
 - **Czechia** and **Poland** – carbon-intensive, centralised economies, highly exposed to EU ETS I & II, with marked variation in perceived local corruption and favouritism (Charron, Lapuente, and Bauhr, [2024](#); Konradt and Mangiante, [2025](#)).

Overview of experimental design

- Respondents evaluate **paired institutional profiles** that vary only along **two orthogonal dimensions**: *temporal rigidity (commitment–flexibility)* and *composition (technocracy–representation)*.
- **Constant across profiles**: policy domain (climate compensation), programme content, and total budget.
- Each respondent completes **choice tasks** and **profile ratings**, with an explicit “**no meaningful difference**” option and confidence measure.
- Main estimands:
 - **AMCEs** (Average Marginal Component Effects),
 - **Marginal Means** (MMs),
 - **Feature Choice Probabilities** (FCPs).
- Analysis pre-registered; randomisation verified via balance checks.

Pre-treatment measures: Temporal exposure

Comprehensive exposure concept: captures both **material and immaterial, individual and regional** dimensions.

Objective present exposure (E_{pres}): carbon-intensive employment (ISCO–NACE), household energy dependence, regional fiscal capacity and investment intensity.

Objective future exposure (E_{fut}): skill distance to green jobs, regional carbon intensity, and share of high-emitting employment.

Subjective exposure: perceived present and future effects on income, employment, energy and transport costs, local status, and community prospects.

Temporal structure index:

$$D = z(E_{\text{pres}}) - z(E_{\text{fut}}) \quad (\text{higher } D = \text{more present-oriented losses})$$

Note: $z(\cdot)$ denotes within-country standardisation of each component ($\text{mean} = 0$, $\text{sd} = 1$) to ensure comparability.

Pre-treatment measures: RITI and complementary covariates

Representation–Impartiality Trade-off Index (RITI)

Measures the perceived trade-off between **incongruence** (distance from decision-makers) and **impartiality** (impersonal administration).

$$RITI = z(\text{Incongruence}) - z(\text{Impartiality}) \quad (\text{higher RITI} = \text{preference for technocracy})$$

Complementary covariates

- **Climate beliefs and knowledge:** support for mitigation, EU Green Deal (Dechezleprêtre et al., 2025; Kollberg et al., 2025)
- **Evaluative logic:** *efficiency* (false positives) vs. *deservingness* (false negatives)
- **Institutional trust:** national/subnational/EU, perceived local corruption
- **Socio-demographics:** education, income, employment sector, region

Note: All components are standardised within country (*mean* = 0, *sd* = 1).

Treatment: Vignette

*"The government is establishing a new **Compensation Commission** to decide who should receive support and how funds should be distributed during the green transition. Below are two possible versions of how this Commission could be organised. Please read each description carefully and indicate which one you would prefer."*

Treatment: Attributes and levels

Dimension	Level A	Level B
Temporal rigidity	Commitment: Eligibility rules and regional allocations are fixed in advance for 4–5 years and cannot be changed during this period. Ensures predictability and credibility.	Flexibility: Rules are reviewed annually and may be adjusted within the total budget to reflect changing needs. Promotes adaptability but lowers predictability.
Composition	Technocratic commission: Expert civil servants appointed for fixed terms, operating at arm's length from parties and interest groups under formal selection criteria.	Descriptively representative commission: Non-electoral body including delegates from trade unions, employers, and regional or sectoral organisations (no politicians). Decisions taken collectively after consultation.

Note: The policy domain (climate compensation), total budget, and funding scope are held constant across profiles to isolate institutional effects.

Treatment: Task structure

- **Task format:** Each respondent completes **six paired-choice tasks**, each comparing two institutional profiles (A vs. B). Profiles vary only along the two institutional dimensions (temporal rigidity and composition); all other programme features are held constant.
- **Design coverage:** The 2×2 factorial design yields four unique institutional profiles: (1) *Commitment–Technocratic*, (2) *Commitment–Representative*, (3) *Flexibility–Technocratic*, and (4) *Flexibility–Representative*. Random pairing across tasks ensures full coverage and orthogonality.
- **Rationale for six tasks:** Provides sufficient within-respondent variation for precise AMCE and interaction estimates (e.g., $D_i \times A_{ijk}$, $\text{RIT} \times A_{ijk}$), while keeping respondent burden moderate. Standard in conjoint designs balancing **precision** and **fatigue** (Cuesta, Egami, and Imai, 2022; Hainmueller, Hopkins, and Yamamoto, 2014).

Outcomes

Primary outcome: Revealed institutional preference

$$Y_{ij} = \begin{cases} 1, & \text{if respondent } i \text{ chooses profile } j, \\ 0, & \text{otherwise.} \end{cases}$$

Binary indicator of institutional choice in the conjoint task.

Secondary outcomes:

- **Policy support:** S_i – post-treatment support for climate policy (e.g. EGD, carbon pricing, bans/standards).
- **Perceived institutional qualities:** C_{ij} (credibility), I_{ij} (impartiality), R_{ij} (representation)
7-point profile ratings capturing perceived procedural features.
- **Confidence and indifference:** K_i (confidence in choice) and N_i (“no meaningful difference”).

Measurement: All outcomes standardised within country ($mean = 0$, $sd = 1$) for comparability.

Estimands and estimation

- **Estimands:**

- **AMCEs:** β_k — average causal effect of attribute k (Hainmueller, Hopkins, and Yamamoto, [2014](#)).
- **Conditional AMCEs:** θ_k — moderation of attribute k 's effect by M_i .
- **Marginal Means (MMs):** $\mathbb{E}[Y_{ij} | A_{ijk} = a]$ (Leeper, Hobolt, and Tilley, [2020](#)).
- **Feature Choice Probabilities (FCPs):** simulated attribute salience measures (Abramson, Kocak, and Magazinnik, [2022](#); Abramson, Kocak, Magazinnik, and Strezhnev, [2023](#)).

- **Estimator:** Linear probability models (LPM) with respondent-level clustered standard errors. Country-specific analyses complemented by pooled models for cross-case comparison.

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- **Specification:** $Y_{ij} = \alpha + \sum_k \beta_k A_{ijk} + \gamma M_i + \sum_k \theta_k (A_{ijk} \times M_i) + \varepsilon_{ij}$

- Y_{ij} : binary profile choice of respondent i in task j ;
- A_{ijk} : level of attribute k (e.g., temporal or compositional dimension);
- M_i : pre-treatment moderator (D_i = exposure or $RITI_i$ = representation).

- **Inference:** In addition to clustered SEs, design-based randomisation inference for robustness.

Parameters and expected signs

Parameter	Interpretation	Moderator	Expected sign
H1: $\theta_{\text{Commitment} \times M}$	Preference for commitment increases with more present exposure (declines for future exposure)	Exposure (D_i)	+ (present losses $\Rightarrow$ commitment)
H2: $\theta_{\text{Representation} \times M}$	Preference for representation rises with exposure to present losses	Exposure (D_i)	+ (affected groups $\Rightarrow$ representation)
H3: $\theta_{\text{Representation} \times M}$	Preference for representation declines with impartiality concern	RITI	– (impartiality $\Rightarrow$ technocracy)
H4: $\theta_{\text{Representation} \times M}$	Among unaffected: efficiency (high RITI) $\Rightarrow$ technocracy; deservingness (low RITI) $\Rightarrow$ representation	RITI	– on average (net technocratic shift)

Note: M_i alternates between exposure (D_i) and institutional beliefs (RITI). **H5** will be tested by regressing post-treatment policy support on institutional design preferences and their alignment with predicted profiles.

Robustness: Conceptual

- **Information equivalence (ex ante):** Both institutional dimensions convey equal informational load and evaluative tone. Ensured through parallel wording, matched text length, and pretesting for perceived specificity and salience.
- **Interpretive clarity:** One practice vignette precedes the conjoint; factual recall and clarity ratings (1–7) follow each block. Misunderstanding recorded when recall fails or clarity < 4; sensitivity analyses compare results with/without flagged cases.
- **Interpretive symmetry:** Attribute wording pretested for neutrality. Identical phrasing used across languages to avoid directional framing.
- **Exclusion of policy confounds (ex post):** Follow-up probes used to confirm respondents perceive constant total budget and a fixed government tier. If placebo AMCEs (Eggers, Tuñón, and Dafoe, [2024](#)) differ from zero, models will be re-estimated conditioning on placebo outcomes (e.g., budget or government control).

Robustness: Estimation

- **Model specification:** Re-estimate AMCEs using logit models to check robustness to link function. Compare coefficients and predicted probabilities with linear probability model (LPM) estimates.
- **Response-quality robustness:** Exclude respondents in the lowest engagement quartile (composite of response variance, and recall accuracy); compare AMCEs and MMs to the full-sample estimates.
- **Learning and fatigue effects:** Interact attribute indicators with task order (Stantcheva, [2023](#)) to detect attenuation or adaptation over rounds. AMCE stability across task index would confirm absence of learning or satisficing (Bansak et al., [2021](#)).
- **Subsample and heterogeneity checks:** Compare AMCEs across exposure (D) and RITI quartiles; test for non-linearity.

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Conclusion and roads not yet taken

- We theorise two **orthogonal institutional trade-offs**: **commitment vs. flexibility** and **technocratic insulation vs. descriptive representation**.
 - The conjoint experiment seeks to test whether **exposure structures** and **representation-related considerations** shape preferences—and, in turn, **support for climate policy**.
- Important, given that institutions for climate compensation are increasingly being set up
- **Caveat**: Recognise that this presentation reflects the **most ambitious version**; will likely have to be scaled down given resource constraints / lack of funding.
- Next steps:
 - Refine and clarify the theoretical framework based on feedback: generate more (empirical) bang for the (theoretical) buck.
 - Minimise complexity of survey.
 - Pre-register and implement the experiment.

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Outline

Appendix

Estimands

Power analysis

Marginal Means (MM): Formal definition

Setup. Two binary attributes: **Temporal design** $K_1 \in \{\text{Commitment } (C), \text{Flexibility } (F)\}$ and **Compositional design** $K_2 \in \{\text{Technocratic } (T), \text{Representative } (R)\}$.

A profile is the pair $(K_1, K_2) \in \{C, F\} \times \{T, R\}$. In each task, respondents choose one of two profiles.

Outcome. $Y = 1$ if the left (or focal) profile is chosen, 0 otherwise.

Marginal Mean (MM) for a level a of attribute K :

$$\text{MM}_K(a) \equiv \Pr(\text{chosen profile has } K = a) = \mathbb{E}_{K' \neq K} \left[\Pr(Y = 1 \mid K = a, K') \right].$$

In words: the *absolute probability* that the chosen profile carries level a , averaging over the randomised distribution of the other attribute(s).

Link to AMCE (binary K). With balanced randomisation:

$$\text{AMCE}_K(a \text{ vs } b) = \text{MM}_K(a) - \text{MM}_K(b), \quad \text{and} \quad \text{MM}_K(a) + \text{MM}_K(b) = 1.$$

Hence $\text{AMCE}_K(a \text{ vs } b) = 2 \text{MM}_K(a) - 1$ (and symmetrically for b).

Feature Choice Probability (FCP): Formal definition

Average Feature Choice Probability (FCP) for a vs. b of the same attribute K :

$$\text{FCP}_K(a \succ b) \equiv \mathbb{E}_{K' \neq K} \left[\Pr \left(\text{choose } (K=a, K') \mid (K=a, K') \text{ vs } (K=b, K') \right) \right].$$

Interpretation: the probability that a wins head-to-head against b when the competing profiles are identical on all other attributes K' , averaged over the randomised distribution of K' .

Properties:

- Symmetric: $\text{FCP}_K(a \succ b) + \text{FCP}_K(b \succ a) = 1$.
- Direct comparison only – excludes indirect contests (unlike AMCE).
- Intuitive interpretation: a “vote share” for one level when all else is equal.
- $\text{FCP} = 0.5$ indicates indifference; above (below) 0.5 indicates preference for a (b).

Illustration: Temporal design (Commitment vs. Flexibility)

Direct head-to-head contests differing only in temporal design:

$$p_{CT,FT} \equiv \Pr(\text{choose } (C, T) \mid (C, T) \text{ vs } (F, T)) = 0.58,$$

$$p_{CR,FR} \equiv \Pr(\text{choose } (C, R) \mid (C, R) \text{ vs } (F, R)) = 0.66.$$

Feature Choice Probability (Commitment $\succ$ Flexibility):

$$\text{FCP}_{K_1}(C \succ F) = \frac{1}{2}(p_{CT,FT} + p_{CR,FR}) = \frac{1}{2}(0.58 + 0.66) = 0.62.$$

Interpretation: When Commitment faces Flexibility holding composition fixed, Commitment wins 62% of direct pairings.

Marginal Means (absolute, not pairwise):

$$\text{MM}_{K_1}(C) = 0.61, \quad \text{MM}_{K_1}(F) = 0.39.$$

Interpretation: Across all contests (composition randomised), 61% of chosen profiles feature Commitment.

AMCE consistency and compositional analogue

Consistency check with AMCE (binary attribute).

$$\text{AMCE}_{K_1}(C \text{ vs } F) = \text{MM}_{K_1}(C) - \text{MM}_{K_1}(F) = 0.61 - 0.39 = 0.22 = 2 \times 0.61 - 1.$$

Interpretation: The AMCE indicates that switching from Flexibility to Commitment increases the probability of choosing the profile by 22 percentage points on average. However, the FCP (0.62) shows that Commitment wins in 62% of direct pairings—similar in direction, but not identical because AMCE also includes *indirect* comparisons.

Compositional design analogue (Representation vs. Technocracy):

$$\text{FCP}_{K_2}(R \succ T) = \frac{1}{2} \left(\Pr((C, R) \text{ wins vs } (C, T)) + \Pr((F, R) \text{ wins vs } (F, T)) \right).$$

Here, $\text{MM}_{K_2}(R)$ is the absolute probability the chosen profile has Representation, averaging over temporal design.

Power analysis: Design and assumptions

- **Objective:** Assess the minimum detectable effect size (MDE) for AMCEs and key interactions ($A_{ijk} \times D_i$, $A_{ijk} \times RITI_i$) under the proposed conjoint design.
- **Simulation framework:** Synthetic data generated using fully randomised attribute levels, with 6 tasks per respondent and binary choices.
- **Assumptions:**
 - $N = 2000$ respondents per country (12,000 choice observations).
 - Balanced attribute levels; profile randomisation fully independent.
 - AMCE variance from prior conjoint studies on institutional design.
 - Clustered standard errors at respondent level.
- **Power target:** 80% power at $\alpha = 0.05$ to detect AMCEs ≥ 0.03 – 0.05 (equivalent to 3–5 percentage-point changes in choice probability).

Power analysis: Simulated results

- **Main effects (AMCEs):** With $N = 2000$, 6 tasks per respondent, and 4 attribute levels, MDE 0.035 (SD 0.018). → High precision for detecting attribute main effects across countries.
- **Moderation effects (Exposure, RITI):** For interactions $A_{ijk} \times M_i$ (where M_i is continuous), MDE 0.05–0.07 depending on variance of M_i . Power rises to >85% for binary subgroup contrasts (e.g., top vs. bottom quartile).
- **Cross-country pooled model:** Pooling 4 countries ($N_{\text{total}} = 8000$) reduces MDEs by 40%, supporting identification of smaller cross-national differences.
- **Robustness:** Simulations repeated for 4–8 tasks per respondent confirm 6-task design optimises precision–fatigue trade-off.

Note: All simulations assume balanced randomisation and linear probability estimation. Simulation code and outputs will be provided with pre-registration materials.